

Helm Charts Tutorial (Beginner) — create, deploy (OpenShift), and store

This is the **hands-on tutorial** part.

If you want the “what is Helm / structure / workflow” overview, read:

- [dataengineering/cloud/helm-charts-intro.md](#)
-

1) Goal (what we will build)

We will:

- create a chart
 - configure it to deploy **nginx**
 - render the YAML (“compile”)
 - deploy it to an OpenShift project
 - learn 3 simple ways to store/share the chart
-

2) Requirements

- Helm installed: `helm version`
 - OpenShift CLI installed: `oc version`
 - Access to a cluster (login credentials)
-

3) Create a starter chart

```
``bash mkdir -p helm-demo && cd helm-demo helm create hello-chart ``
```

Chart structure reminder:

Helm chart structure (what's inside a chart folder)

```
mychart/
  Chart.yaml      (name/version)
  values.yaml     (your settings)
  templates/      (YAML templates)
    deployment.yaml
    service.yaml
    route.yaml    (OpenShift)
  charts/         (optional deps)
```

4) Configure the app to deploy (values)

Edit `hello-chart/values.yaml` and set a real image. Example:

```
```yaml
replicaCount: 1

image: repository: nginx tag: "1.25"

service: type: ClusterIP port: 80 ```
```

---

### 5) Validate the chart

```
```bash
helm lint ./hello-chart ```
```

6) Render (“compile”) to YAML locally

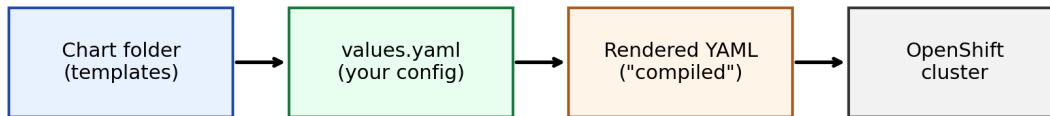
This is the safest step before you touch the cluster:

```
```bash
helm template hello ./hello-chart ```
```

If the output YAML looks wrong, fix `values.yaml` or `templates` and rerun.

Workflow reminder:

Helm workflow: chart → rendered YAML → cluster



Helm does two jobs: 1) render templates to final YAML 2) apply that YAML to the cluster + store a release record

---

## 7) Deploy to OpenShift

### Step A — Login and choose a project

```
``bash oc login https://api.:6443 oc new-project hello-project
```

**or**

```
oc project hello-project ``
```

### Step B — Install the chart into that namespace

```
``bash helm install hello ./hello-chart -n hello-project ``
```

Verify:

```
``bash helm list -n hello-project oc get all -n hello-project ``
```

### Step C — Upgrade (after changing values/templates)

```
``bash helm upgrade hello ./hello-chart -n hello-project ``
```

### Step D — Uninstall

```
``bash helm uninstall hello -n hello-project ``
```

---

## 8) OpenShift routing: Ingress vs Route (beginner view)

- Kubernetes often uses **Ingress**.
- OpenShift commonly uses **Route**.

If you want a URL on OpenShift, you may add a Route template.

Example hello-chart/templates/route.yaml:

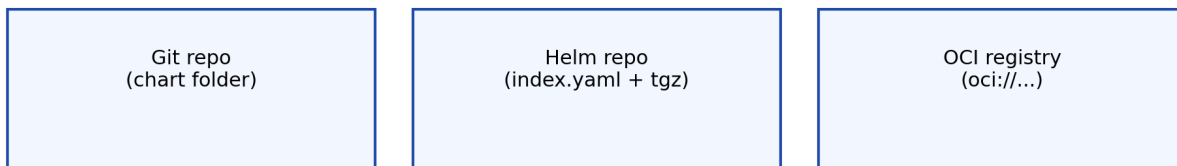
```
``yaml apiVersion: route.openshift.io/v1 kind: Route metadata: name: {{
include "hello-chart.fullname" . }} spec: to: kind: Service name: {{
include "hello-chart.fullname" . }} port: targetPort: http ``
```

Tip: teams usually make this optional with a value like `route.enabled: true`.

---

## 9) How to store/share your chart (3 easy options)

How teams store/share Helm charts (3 common ways)



CI/CD typically runs: `helm upgrade --install ...` (from Git, repo, or OCI)

### Option 1 — Store chart folder in Git (simplest)

```
``bash helm upgrade --install hello ./hello-chart -n hello-project -f
values-prod.yaml ``
```

### Option 2 — Helm repo (package + index)

```
``bash helm package ./hello-chart helm repo index . ``
```

Host the `.tgz` + `index.yaml` (GitHub Pages, S3, Nexus/Artifactory).

### Option 3 — OCI registry (modern)

High-level flow:

```
```bash helm package ./hello-chart helm registry login helm push hello-chart-0.1.0.tgz oci:///charts
helm install hello oci:///charts/hello-chart -n hello-project ```
```

10) Final mental model

- Chart = blueprint
- Values = settings
- Render (“compile”) = blueprint → final YAML
- Install/upgrade = apply YAML to OpenShift
- Store/share = Git or Helm repo or OCI registry